Federico Scaramuzza

PhD candidate — Chronobiology and Computational Biology

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EDUCATION

11-2021 – present

PhD, Molecular Biology, VBC PhD program
University of Vienna, Department of Neuroscience and Developmental Biology, Vienna, Austria
Project: Molecular mechanisms of circadian plasticity in response to moonlight and the circalunar clock in a marine annelid.
Supervisor: Prof. Kristin Tessmar-Raible. Expected defence: June 2027

2019 – 2021

M.Sc. Molecular Biology — 110/110 *cum laude*
University of Padova, Department of Biology, Padova, Italy
Thesis: Circadian dysfunction and neurodegenerative disorders: *Drosophila* as a model to investigate the link between rhythmic mitochondrial dynamics and neurodegeneration.
Supervisor: Prof. Gabriella Margherita Mazzotta.

2016 – 2019

B.Sc. Molecular Biology — 102/110
University of Padova, Department of Biology, Padova, Italy
Thesis: Role of the clock gene *Period3* in the human cell-autonomous clock.
Supervisor: Prof. Rodolfo Costa.

PUBLICATIONS

Peer-reviewed

1. Mat A.M., **Scaramuzza F.**, Ćorić A., Klopp C., Matabos M., Tessmar-Raible K. (2026). Endogenous rhythms in deep-sea mussel cell culture and molecular clock characterization. *EMBO Reports*. Accepted for publication, Journal cover choice
2. Andreatta G., **Scaramuzza F.**, Ćorić A., Orel L., Mat A.M., Takekata H., Poehn B., Milivojev N., Tessmar-Raible K. (2026). Photoreceptor control of *Platynereis* growth and lifespan via evolutionarily conserved molecular pathways. *PNAS*. https://doi.org/10.1073/pnas.2514719123
3. Milivojev N., **Scaramuzza F.**, Brum P.O., Velastegui Gamboa C.L., Andreatta G., Raible F., Tessmar-Raible K. (2025). Light-modulated stem cells in the camera-type eye of an annelid model for adult brain plasticity. *Nature Communications*. https://doi.org/10.1038/s41467-025-65631-0

Preprints

1. **Scaramuzza F.**, Andreatta G., Mat A.M., Navid A., Poehn B., Zurl M., Orel L., Brum P.O., Herzel H., Tessmar-Raible K. A circalunar oscillator and moonlight tune the circadian clock to moon phase via *cwo/Dec1&2*. *bioRxiv*. https://doi.org/10.64898/2026.10.14.757926
2. Brum P.O., Abdi A., **Scaramuzza F.**, Kepcija T., Hastings M.H., Loudon A., Tessmar-Raible K., Urban Avellaneda N. (2026). PER2- and state-dependent transcriptional programs gate neural stem cell proliferation with niche-specific circadian autonomy. *bioRxiv*. https://doi.org/10.64898/2026.09.14.751519

In preparation

1. Ćorić A., Wulf P.O., Vu H.H., Orel L., Poehn B., Krishnan S., **Scaramuzza F.**, Andreatta G., Wolf E., Tessmar-Raible K. Towards a molecular understanding of moon-controlled timers: Linking the light valence detector L-Cry to the circalunar clock in the marine annelid *Platynereis dumerilii*.
2. Adelmann L., Panayotova M., **Scaramuzza F.**, Raible F. Arachidonic Acid acts as a booster of regeneration competence in *Platynereis dumerilii*.

TALKS

09-2026 — **Short talk: Tuning the circadian clock to the moon - a candidate integrator of circadian and circalunar timing in an annelid model**
ICYMARE 2026, Bremen, Germany

06-2026 — **Invited talk: Decoding the Molecular Integration of Solar and Lunar Rhythms in Biological Clock Plasticity**
Euro Evo Devo, Glasgow, United Kingdom

03-2026 — **Short talk: Molecular mechanisms underlying circadian–circalunidian clock plasticity**
EMBO|EMBL Symposium: Biological Oscillators — Rhythms and Synchronisation across Scales, Heidelberg, Germany

POSTER PRESENTATIONS

08-2025 — **Crosstalk of sun and moon: mechanisms underlying solar and lunar rhythm integration in *Platynereis dumerilii***
European Biological Rhythms Society (EBRS), Lübeck, Germany

10-2022 — **The relationship between solar- and lunar-controlled rhythms in *Platynereis dumerilii***
EBRS Chronobiology School, Munich, Germany

RESEARCH EXPERIENCE

11-2021 – present — **PhD, Molecular Biology**
University of Vienna, Department of Neuroscience and Developmental Biology, Vienna, Austria
Using the marine bristleworm *Platynereis dumerilii* to investigate how moonlight and the circalunar clock shape circadian plasticity, with the aim of identifying the underlying molecular mediators. My PhD project combines computational modelling of coupled oscillators using ordinary differential equations (ODEs) with experimental approaches, including dual-luciferase reporter assays, immunohistochemistry, hybridisation chain reaction (HCR), western blotting, bulk RNA sequencing, and behavioural tracking.

01-2021 – 09-2021

M.Sc. research project
Costa lab, University of Padova, Department of Biology, Padova, Italy
Investigated the effects of circadian disruption on mitochondrial dynamics and neurodegeneration in *Drosophila melanogaster*. Combined mutant and transgenic fly lines with qPCRs, western blotting, confocal microscopy, transmission electron microscopy, and high-resolution respirometry (Oroboros) to link cellular phenotypes with metabolic alterations.

10-2020 – 01-2021

Research assistant
Costa lab, University of Padova, Department of Biology, Padova, Italy
Investigated the interaction between the circadian photoreceptor Cryptochrome (CRY) and Calmodulin (CaM) in *D. melanogaster*. Tested mutant CRY variants using yeast two-hybrid assays and analysed tagged CRY and CaM overexpressed in S2 cells using co-immunoprecipitation and western blotting.

01-2019 – 07-2019

B.Sc. research project
Costa lab, University of Padova, Department of Biology, Padova, Italy
Investigated the mitochondrial uncoupling protein gene *ucp4c* in *Drosophila suzukii* as a potential target for population control. Characterised its rhythmic transcript expression using qPCR and designed a CRISPR–Cas9 loss-of-function mutant.
Compilation thesis: "Role of the clock gene *Period3* in the human cell-autonomous clock"

TEACHING AND OUTREACH

2026

Symposium organiser
Vienna BioCenter PhD Symposium 2026
Co-organised the Vienna BioCenter PhD Symposium 2026.

2026

Session organiser and host
ICYMARE 2026
Organised and hosted Session 4.1 "Temporal Organisation in Marine Life Across Biological Scales: from Molecules to Ecosystems".

2026

Peer review
Nature Communications
Reviewed research papers as part of the Nature Communications' Early Career Researchers Co-Review Initiative.

2021 – present

Student supervision
University of Vienna, Department of Neuroscience and Developmental Biology, Vienna, Austria Vienna
Barbara Stauber (BSc), Feray Feuerhake (internship), Estefania Isabella Leth (BSc), Mirela Panayotova (BSc), Anita Navid (MSc), Sebastian Riegler (BSc), Nikola Edelsbrunner (internship), Iona Elisa Bergerhoff (internship), Lucia Allmendinger (internship).

FELLOWSHIPS AND AWARDS

2020

Erasmus+ mobility grant
University of Padova, Italy
Awarded; not taken up owing to COVID-19 travel restrictions.

SKILLS

Wet lab CRISPR-Cas9 mutagenesis; Molecular cloning; Dual-luciferase reporter assays; Western Blot; Immunohistochemistry; *in situ* Hybridisation Chain Reaction (HCR); Confocal and super-resolution imaging (LSM980, LSM900, LSM700); RNA extraction; Genotyping; Husbandry and crossing of *Platynereis dumerilii* and *Drosophila* spp.; Tissue Culture.

Computational Bulk and single-cell RNA-seq data processing and analysis (Trimmomatic, Salmon, STAR, Cell Ranger, Subread, DESeq2, Seurat); Rhythmicity and differential-rhythmicity analysis (RAIN, dryR); Comparative genomics and genome annotation; Modelling of coupled oscillators with ordinary differential equations (ODEs); HPC job management (SLURM); Public data deposition (ENA); Variant calling (nf-core/sarek).

Software R; Python; Linux; Bash scripting; Git; LaTeX; Fiji/ImageJ; CLC Main Workbench; Adobe Suite; GIMP; Inkscape.

Languages Italian (native); English (fluent, IELTS 8.0); German (B1, Sprachenzentrum Wien).

ADDITIONAL EDUCATION

2015 – 2020

Introductory Diploma in Opera Singing — 9/10
Conservatory of Adria "A. Buzzolla", Adria (RO), Italy
Studied with Mº Roberto Scaltriti and Mº Claudio Desderi. Main roles include Figaro in *Le nozze di Figaro* (Adria city theatre, 2019) and Don Giovanni in *Il dissoluto punito, ossia il Don Giovanni* (Perugia, 2019) by W.A. Mozart.

2008 – 2019

M.Mus. Classical Guitar — 9/10
Conservatory of Adria "A. Buzzolla", Adria (RO), Italy
Completed advanced training in classical guitar performance, interpretation, and chamber music. Performed regularly as a soloist and chamber musician, developing extensive experience in ensemble collaboration and public performance.
Supervisor: Mº Alberto La Rocca.

REFERENCES

Prof. Kristin Tessmar-Raible
Department of Neuroscience and Developmental Biology, University of Vienna, Austria
kristin.tessmar-raible@univie.ac.at

Ass. Prof. Florian Raible
Department of Neuroscience and Developmental Biology, University of Vienna, Austria
florian.raible@univie.ac.at

Prof. Hanspeter Herzel
Humboldt-Universität zu Berlin, Germany
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Prof. Charlotte Helfrich-Förster
Neurobiology and Genetics, Theodor-Boveri Institute, Biocenter, University of Würzburg, Germany
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